

A. Stakeholder Engagement Plan

Project Title: SuperBPE Tokenizer for Low-Resource African Languages

Focus Area: Preprocessing Infrastructure, NLP Equity, and Sub-word Tokenization

Vocabulary Limit: 20k Tokens (SuperBPE + BPE) across Representative Language Families

1. Identification of Stakeholder Groups

Group 1: African AI Developers & Local Tech Talent

- **Who They Are:** Software engineers, machine learning practitioners, university researchers, and tech youth across Africa building downstream AI applications (e.g., translation, voice tools, chatbots).
- **Why They Are Important:** They are the primary technical end-users of the tokenizer. If local developers cannot easily integrate this tokenizer into their ML pipelines (e.g., via Hugging Face), the project fails its core deployment and utility objectives.
- **What Matters Most to Them:** Low computational overhead, reduced sequence length (token reduction), low memory footprint, API cost savings, clear documentation, and open-source library compatibility.

Group 2: Native Speakers & Grassroots Linguistic Communities

- **Who They Are:** Native-speaking community members, cultural preservers, and grassroots African NLP collectives (e.g., Masakhane contributors) across selected African language families.
- **Why They Are Important:** They possess ground-truth cultural and linguistic knowledge. Their involvement ensures that sub-word segmentation respects native morphology and preserves diacritical and tonal nuances, preventing cultural erasure.
- **What Matters Most to Them:** Cultural preservation, accurate representation of native accents and diacritical marks ("ticks"), linguistic integrity, protection against bias, and digital inclusion without Westernized approximations.

2. Potential Benefits and Risks

Stakeholder Group	Potential Benefits	Potential Harms / Concerns	Ethical, Social, Cultural & Economic Considerations
African AI Developers & Tech Talent	Drastically reduces token inflation (from 3--10 × down to optimal sub-word segmentation), cutting training/inference compute requirements, memory overhead, and operational API costs. Enables low-cost model training on modest local hardware.	Risk of integration friction if custom SuperBPE vocabulary formats are incompatible with standard NLP libraries, or if out-of-domain vocabulary causes unexpected runtime errors.	Economic & Social: Democratizes AI access by removing prohibitive compute cost barriers, empowering Africa's expanding youth workforce to actively participate in the global AI economy.

Stakeholder Group	Potential Benefits	Potential Harms / Concerns	Ethical, Social, Cultural & Economic Considerations
Native Speakers & Grassroots Communities	Dignified digital representation that accurately preserves orthography, tonal markers, and special characters. Facilitates accessible downstream localized software and educational tools.	Misrepresentation or corruption of native meanings if BPE over-segments root words into nonsensical byte sequences; higher-resource African languages overshadowing lower-resource dialects within the 20k token vocabulary limit.	Cultural & Ethical: Respecting data sovereignty, preventing extractive data practices, and ensuring digital inclusion respects authentic linguistic identity without stripping cultural context.

3. Engagement Method (Action Catalogue Alignment)

Method for Group 1: Participatory Design (from Action Catalogue)

- **Why Appropriate & Suitable:** Developers are hands-on technical practitioners who require active co-creation rather than passive consultation. Participatory Design directly involves end-users in testing software interfaces and tokenization performance in real-time.
- **How Engagement Will Be Conducted:** We will conduct collaborative virtual hackathons and technical co-design sprints. Developers will test prototype SuperBPE tokenizers on their own datasets, benchmark token compression ratios against standard tokenizers, report integration bugs via GitHub, and co-design Python/Hugging Face wrapper APIs.

Method for Group 2: User Committees (from Action Catalogue)

- **Why Appropriate & Suitable:** Native speakers need a structured, ongoing governance role to oversee linguistic accuracy across multiple language families without feeling exploited or briefly surveyed.

- **How Engagement Will Be Conducted:** We will get native speakers to audit our dataset during dataset curation and vocabulary allocation phases to review segmented word lists, evaluate diacritic preservation, audit out-of-domain vocabulary samples, and approve token dictionary releases.

4. Influence on Project Design

- **Dataset Curation:** Feedback from User Committees will identify missing sub-family text spans, correct bad text formatting in news/bible corpora, and ensure proper diacritic representation across target languages.
- **Model Design & Algorithms:** Developer feedback during Participatory Design sessions will directly tune the SuperBPE sub-word merging heuristics, while User Committee feedback will guide the selection of temperature-based data sampling (α -parameter values) to prevent high-resource language dominance within the 20k token constraint.
- **Outputs & Vocabulary Indexing:** Audits by native speakers will ensure that token string outputs correspond to meaningful morphological roots rather than arbitrary byte splits.
- **Evaluation Frameworks:** Evaluation will expand beyond raw loss metrics to include token compression ratio (evaluating cost savings for developers) and linguistic fidelity scores (evaluated by User Committees).
- **Deployment Decisions:** Developer input will dictate containerization formats, Hugging Face Hub integration scripts, and lightweight library wrappers to ensure seamless integration into downstream model pipelines.

B. Which 2 stakeholders answer is at the top

C. By integrating **Participatory Design** (with African AI developers) and **Native Speakers Audit** (with native-speaking language preservers), your team actively addresses key ethical, technical, and social imperatives necessary to build a responsible AI system.

1. Improving Trust

- **Participatory Design:** Involving local developers directly in technical co-design sprints and open-source testing builds radical transparency. Developers see how SuperBPE processes sub-word merges in real time rather than relying on a closed, black-box pipeline.
- **User Committees:** Establishing an ongoing advisory board gives native communities governance over their language data. When language speakers see that their feedback

directly dictates token boundaries, trust shifts from skepticism to institutional co-ownership.

2. Identifying Risks

- **Linguistic Misrepresentation:** Purely data-driven BPE algorithms can blindly chop meaningful morphological roots or strip diacritical marks ("ticks"). User Committees act as human guardrails, spotting orthographic corruption and tonal misinterpretations early.
- **Integration Failures:** Participatory Design allows developers to surface workflow bugs, memory leaks, and library incompatibilities on actual resource-constrained hardware before public deployment.

3. Increasing Representation

- **Dialectal & Family Inclusion:** Standard tokenizers skew heavily toward dominant global languages like English or French, leading to severe token inflation for African text. Engaging representatives across representative language families ensures underrepresented dialects and low-resource sub-families are actively advocated for.
- **Grassroots Co-creation:** Combining developer and native-speaker groups ensures both technical talent and non-technical community elders have an equal voice in shaping AI infrastructure.

4. Refining the Dataset

- **Data Quality over Quantity:** User Committees audit training corpora to flag corrupted Optical Character Recognition (OCR) outputs, improper character encodings, or skewed bible/news text sources, preserving authentic language structures.
- **Vocabulary Allocation Tuning:** Stakeholder feedback directly informs temperature-based sampling (α -parameter tuning), preventing higher-resource African languages from dominating the limited 20k token vocabulary capacity.

5. Improving Model Outputs

- **Lowering Token Inflation:** Co-designed sub-word heuristics ensure token outputs retain semantic, morphological meaning rather than arbitrary byte fragments. This cuts down token length by 3--10 $\times$, lowering compute requirements and inference costs for downstream LLMs.
- **Preserving Diacritical Integrity:** Direct validation by native speakers ensures downstream models trained on these tokens maintain high fidelity in translation, text-to-speech, and sentiment analysis tasks without erasing cultural nuances.

D. Continuous, structured stakeholder engagement ensures that the **SuperBPE Tokenizer** transitions from a technical prototype into a long-term, community-owned piece of critical AI infrastructure. By engaging both African AI developers and native-speaking communities throughout the project lifecycle, the project systematically builds **fairness, transparency, accountability, and community ownership**:

1. Fairness

- **Language Equity across Families:** Unchecked sub-word algorithms naturally prioritize dominant, higher-resource languages, leading to severe token inflation for underrepresented dialects. By consulting native speakers across distinct linguistic families and sub-families, the team can refine temperature-based sampling (α -tuning) to ensure that the 20k token vocabulary capacity is distributed equitably rather than biased toward high-resource languages.
- **Economic Inclusivity:** Engaging local developers ensures the tokenizer directly addresses real-world compute constraints. Lowering token sequences by 3--10 $\times$ reduces compute overhead, allowing developers in resource-constrained regions to build downstream AI tools without prohibitive API and infrastructure costs.

2. Transparency

- **Open Technical Co-Design:** Through *Participatory Design* hackathons and virtual sprints, developers gain full visibility into how sub-word merge decisions are executed. Rather than relying on a black-box tokenizer, open-source code and clear documentation foster deep institutional trust.
- **Linguistic Auditing:** Regular reviews with the *User Committee* (Linguistic Advisory Board) make the vocabulary selection process transparent. Native speakers can trace how character encodings, diacritical marks ("ticks"), and special accents are processed within the SuperBPE dictionary.

3. Accountability

- **Human Guardrails for Algorithmic Errors:** Standard Byte-Pair Encoding can over-segment morphological root words into arbitrary, meaningless byte splits. Ongoing community validation creates a direct accountability loop, enabling the team to catch orthographic corruption and tonal misinterpretations early.
- **Responsible Data Stewardship:** Engaging local language preservers ensures training text corpora (e.g., news, literature, translated texts) are sourced ethically, respecting data sovereignty and avoiding extractive practices.

4. Community Ownership & Long-Term Adoption

- **From Beneficiaries to Co-Creators:** Establishing a permanent *User Committee* shifts native communities from passive data sources to active governors of their digital language representation. This ongoing governance fosters cultural pride and long-term stewardship.

- **Sustained Ecosystem Growth:** When African developers actively participate in building and testing Python wrapper APIs and Hugging Face integrations, they become advocates for the tool. This grassroots adoption ensures the tokenizer becomes a standard foundation for future localized translation tools, educational platforms, and AI applications driven by Africa's expanding youth workforce.